

1. $u_1(x) = 4x^4$, $u_2(x) = 3x^2 - 3x^4$, $u_3(x) = 2x^4 - 5x^2$ are solutions of a second order, homogeneous equation

$$y'' + p(x)y' + q(x)y = 0.$$

What is the general solution of the equation?

- (a) $y = C_1(4x^4) + C_2(3x^2 - 3x^4) + C_3(2x^4 - 5x^2)$
 - (b) $y = C_1x^4 + C_2x^2$
 - (c) $y = C_1x^2 + C_2x^4 + 3x^2 + 4x^4$
 - (d) $y = C_1x^2 + C_2x^4 + C_3$
 - (e) None of the above.
2. Continuing with Problem 1, find the differential equation.
- (a) $y'' + \frac{7}{x}y' + \frac{8}{x^2}y = 0$
 - (b) $x^2y'' - 5xy' - 8y = 0$
 - (c) $y'' - \frac{5}{x}y' + \frac{8}{x^2}y = 0$
 - (d) $x^2y'' - 6xy' + 8y = 0$
 - (e) None of the above.
3. $y'' - \frac{1}{x}y' - \frac{15}{x^2}y = 0$ has solutions of the form $y = x^r$. Find a fundamental set of solutions and give their Wronskian.
- (a) $\{x^{-3}, x^5\}$; $W = 8x$
 - (b) $\{x^{-5}, x^3\}$; $W = 8x^{-3}$
 - (c) $\{x^3, x^5\}$; $W = 2x^7$
 - (d) $\{e^{-5x}, e^{3x}\}$; $W = 2e^{-2x}$
 - (e) None of the above.
4. $y = x^3$ is a solution of $y'' - \frac{5}{x}y' + \frac{9}{x^2}y = 0$. Find a fundamental set of solutions of the equation. (Hint: See Exercises 3.2)
- (a) $\{x^3, x \ln x^3\}$
 - (b) $\{x^3, x^{-3}\}$
 - (c) $\{x^3, x \ln x\}$
 - (d) $\{x^3, x^3 \ln x\}$
 - (e) None of the above.

5. A solution basis for $y'' - 4y' - 12y = 0$ is:

- (a) $\{y_1 = e^{-6x}, y_2 = e^{2x}\}$
- (b) $\{y_1 = e^{4x}, y_2 = e^{-3x}\}$
- (c) $\{y_1 = e^{-4x}, y_2 = e^{3x}\}$
- (d) $\{y_1 = e^{6x}, y_2 = e^{-2x}\}$
- (e) None of the above.

6. The general solution of $y'' + 3y' - 10y = 0$ is:

- (a) $y = C_1e^{-5x} + C_2e^{2x}$
- (b) $y = C_1e^{-5x} + C_2e^{-2x}$
- (c) $y = C_1e^{5x} + C_2e^{-2x}$
- (d) $y = C_1e^{5x} + C_2e^{2x}$
- (e) None of the above.

7. The general solution of $y'' + 2y' + 10y = 0$ is:

- (a) $y = C_1e^{3x} \cos x + C_2e^{3x} \sin x$
- (b) $y = C_1e^{-5x} + C_2e^{2x}$
- (c) $y = C_1e^{-x} \cos 3x + C_2e^{-x} \sin 3x$
- (d) $y = C_1e^{5x} + C_2e^{-2x}$
- (e) None of the above.

8. A solution basis for the differential equation $y'' + 14y' + 49y = 0$ is:

- (a) $\{y_1 = e^{7x}, y_2 = e^{-7x}\}$
- (b) $\{y_1 = e^{-7x}, y_2 = xe^{-7x}\}$
- (c) $\{y_1 = e^{-7x}, y_2 = x\}$
- (d) $\{y_1 = e^{7x}, y_2 = xe^{7x}\}$
- (e) None of the above.

9. The solution of the initial-value problem $y'' - 8y' + 16y = 0$, $y(0) = 1$, $y'(0) = 2$ is:

- (a) $y = e^{-4x} - 6xe^{-4x}$
- (b) $y = \frac{1}{2}e^{-4x} + \frac{3}{2}e^{4x}$
- (c) $y = e^{4x} + 6xe^{4x}$
- (d) $y = e^{4x} - 2xe^{4x}$
- (e) None of the above.

10. A solution basis for the equation

$$y'' + 16y = 0$$

is:

- (a) $\{y_1 = e^{4x}, y_2 = e^{-4x}\}$
- (b) $\{y_1 = \cos 4x, y_2 = \sin 4x\}$
- (c) $\{y_1 = 0, y_2 = e^{4x}\}$
- (d) $\{y_1 = 1, y_2 = e^{-4x}\}$
- (e) None of the above.

11. A solution basis for the equation

$$y'' + 4y' = 0$$

is:

- (a) $\{y_1 = 1, y_2 = e^{-4x}\}$
- (b) $\{y_1 = 1, y_2 = e^{4x}\}$
- (c) $\{y_1 = 0, y_2 = e^{4x}\}$
- (d) $\{y_1 = 1, y_2 = e^{-2x}\}$
- (e) None of the above.

12. The general solution of $y'' + 4y' + 20y = 0$ is:

- (a) $y = C_1 e^{2x} \cos 4x + C_2 e^{2x} \sin 4x$
- (b) $y = C_1 e^{4x} \cos 2x + C_2 e^{4x} \sin 2x$
- (c) $y = C_1 e^{-2x} \sin 4x - C_2 e^{-2x} \cos 4x$
- (d) $y = C_1 e^{-4x} \sin 2x - C_2 e^{-4x} \cos 2x$
- (e) None of the above.

13. $y = C_1 e^{-x} + C_2 e^{5x}$ is the general solution of a second order, linear, homogeneous differential equation. The equation is:

- (a) $y'' - 4y' + 5y = 0$
- (b) $y'' - 4y' - 5y = 0$
- (c) $y'' - 6y' + 5y = 0$
- (d) $y'' - 6y' - 5y = 0$
- (e) None of the above.

14. $y = 3e^{-2x} - 5xe^{-2x}$ is a solution of a second order, linear, homogeneous differential equation.
The equation is:

- (a) $y'' + 4y' = 0$
- (b) $y'' + 4y' + 4y = 0$
- (c) $y'' - 4y' + 4y = 0$
- (d) $y'' - 4y = 0$
- (e) None of the above.

15. $y = -4e^{-3x} \sin x$ is a solution of a second order, linear, homogeneous differential equation.
The equation is:

- (a) $y'' - 6y' + 10y = 0$
- (b) $y'' - 2y' + 10y = 0$
- (c) $y'' + 2y' - 10y = 0$
- (d) $y'' + 6y' + 10y = 0$
- (e) None of the above.

16. $y = 3e^{2x} - 4xe^{-3x}$ is a solution of a linear differential equation with constant coefficients.
The equation is:

- (a) $y'' + y' - 6y = 0$
- (b) $y'' - y' - 6y = 0$
- (c) $y'' + 6y' + 9y = 0$
- (d) $y'' - 3y' + 18y = 0$
- (e) None of the above.

17. Given the linear operator

$$L[y] = y'' - \frac{5}{x}y' + \frac{8}{x^2}y.$$

Calculate $L[4x^2 - 2x^3]$.

- (a) $-3x$
- (b) 4
- (c) $2x$
- (d) $6x^2$
- (e) None of the above.

18. Refer to Problem 17. A fundamental set of solutions of $L[y] = 0$ is:

- (a) $\{x^2, x^4\}$
- (b) $\{x^{-2}, x^6\}$
- (c) $\{x^2, x^{-4}\}$
- (d) $\{x^{-2}, x^{-4}\}$
- (e) None of the above.